AI & ML by Aditya Jain sir --> Lecture 2

HW

```
# 1. What will be printed?

# print(10)

# A. Ten

# B. "10"

# C. 10

# D. Error

Ans: C
```

```
print(10)
```

```
10
```

```
# 2. Guess the output:

# print(5 + 3)

# Output?

# A. 53

# B. 8

# C. 5+3

# D. Error

Ans: B
```

```
print(5+3)
```

```
8
```

```
print("5+3")
```

```
5+3
```

```
print("Python")
print("SQL")

Output?

A.

Python SQL

B.

Python
SQL

C.

SQL
Python

D. Error
```

```
print("Python")
print("SQL")
```

```
Python
SQL
```

```
# Q- What's the output?
Print("Hello")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_784/2674448011.py in <cell line: 0>()
      1 # Q- What's the output?
----> 2 Print("Hello")

NameError: name 'Print' is not defined
```

Next steps: Explain error

```
print("A")
print("B")
print("C")
```

A.

ABC

B.

A B C

C.

A
B
C

D. Error

```
print("A")
print("B")
print("C")
```

```
A
B
C
```

```
print("A", "B", "C")
```

```
A B C
```

Comments

```
This code prints AJ on the screen
print("AJ")
```

```
  File "/tmp/ipykernel_784/2191621304.py", line 1
    This code prints AJ on the screen
         ^
SyntaxError: invalid syntax
```

Next steps: Explain error

```
#This code prints AJ on the screen
print("AJ")
```

```
AJ
```

```
# This code prints AJ on the screen
print("AJ")
```

```
# This code
# prints AJ on the screen

print("AJ")
```

```
# print("Hello")
print("Python")

A.

Hello
Python

B.

Hello

C.

Python

D. Error

Ans: C
```

```
# print("Hello")
print("Python")
```

```
Python
```

Variables

```
name = "AJ"
city = "Nagpur"
id = 24
```

```
print(name)
```

```
AJ
```

```
print(id)
```

```
24
```

```
print(City)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_784/2873433310.py in <cell line: 0>()
----> 1 print(City)

NameError: name 'City' is not defined
```

Next steps: Explain error

```
print(city)
```

```
Nagpur
```

```
#Variable naming rules

# 1. Start: Alphabet , _
# 2. Middle or end --> Alphabet, _, digits
```

```
# student_name --> yes
# _marks --> yes
# City --> yes
# State5 --> yes
# 4city --> no
# student-name --> no
```

```
# variable_name = value
```

```
name = "AJ"
city = "Nagpur"
state = "Maharashtra"
id = 24
```

```
# first name = "Aditya"

firstName = "Aditya"
first_name = "Aditya"
```

```
college = "RCOEM"
college = "IIT Bombay"
print(college)
```

```
IIT Bombay
```

```
college = "RCOEM"
print(college)
college = "IIT Bombay"
print(college)
```

```
RCOEM
IIT Bombay
```

Data Types

```
1. Integer --> int
eg: 50, 4, -6, 0, 22

2. Float --> decimal (also called double)
eg: 5.7, -4.5, 3.6, 2.0

3. String (text data) --> str
 "Aditya", "Nagpur" ,  "Aditya Jain"

4. Boolean --> True/False --> bool
```

```
# "4" --> str
# 5 --> int
# True --> bool
# length = 6.7 --> float
```

```
type("4")
```

```
str
```

```
type(5)
```

```
int
```

```
type(True)
```

```
bool
```

```
type(6.7)
```

```
float
```

```
type("5.7")
```

```
str
```

```
type("True")
```

```
str
```

```
type(false)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_784/27329152.py in <cell line: 0>()
----> 1 type(false)

NameError: name 'false' is not defined
```

Next steps: Explain error

```
type(False)
```

```
bool
```

Taking input from the user

```
name = "Aditya"
```

```
# input()
```

```
name = input()
```

```
aj
```

```
print(name)
```

```
aj
```

```
name = input("Enter your name:")
```

```
Enter your name:AJ
```

```
print(name)
```

```
AJ
```

```
# Q. Take two int as input from the user and print their sum
```

```
a = input()
```

```
4
```

```
b = input()
```

```
5
```

```
print(a+b)
```

```
45
```

```
#string concatenation
```

```
4+5
```

```
9
```

```
"4" + "5"
```

```
'45'
```

```
"a" + "b"
```

```
'ab'
```

```
# input() --> it is always taken as string
```

Type Casting

```
age = "20"
new_age = int(age) #typecasting
type(new_age)
```

```
int
```

```
type(age)
```

```
str
```

```
a = input()
b = input()
c = int(a) + int(b)
print(c)
```

```
4
5
9
```

```
#mentos zindagi
a = int(input())
b = int(input())
c = a+b
print(c)
```

```
5
3
8
```

```
# Q. Take year of birth as input and print the age.
```

```
birth_year = input()
```

```
2005
```

```
age = 2026-birth_year
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_784/2177837109.py in <cell line: 0>()
----> 1 age = 2026-birth_year

TypeError: unsupported operand type(s) for -: 'int' and 'str'
```

Next steps: Explain error

```
birth_year = int(input())
age = 2026-birth_year
print(age)
```

```
2005
21
```

```
# Q. Take length and breadth of the rectangle as input and print its area.
# area = l*b
```

```
l = int(input())
b = int(input())
area = l*b
print(area)
```

```
4
6
24
```

```
l = int(input())
b = int(input())
area = l*b
print(area)
```

```
4.5
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_784/1166909970.py in <cell line: 0>()
----> 1 l = int(input())
      2 b = int(input())
      3 area = l*b
      4 print(area)

ValueError: invalid literal for int() with base 10: '4.5'
```

Next steps: Explain error

```
float("4.5")
```

```
4.5
```

```
l = float(input())
b = float(input())
area = l*b
print(area)
```

```
4.5
2
9.0
```

```
# Q. Take the name, age and height (in inches) of the student as input and print them.
```

```
name = input()
age = int(input())
height = float(input())

print("--------------------------")
print(name, age, height)
```

```
Raj
22
65.5
--------------------------
Raj 22 65.5
```

Quizes

```
Q1

What is a variable?

A. Function
```

```
B. Container for storing data

C. Operator

D. Comment

Ans: B
```

```
Q2

Which is a valid variable?

A- student_name
B- 1student
C- student-name
D- student name

Ans: A
```

```
Q3

Data type of:

type(100) --> int

type("4.5") --> str
```

```
Q4.
Output?

num1 = input()
num2 = input()

print(num1 + num2)

Input:

10
20

Ans: 1020
```

```
num1 = input()
num2 = input()

print(num1 + num2)
```

```
10
20
1020
```

```
Q5 - Predict the Output:


is_student = True
print(type(is_student))

A.

<class 'int'>

B.

<class 'bool'>

C.

<class 'str'>

D. Error

Ans: B
```

```
is_student = True
type(is_student)
```

```
bool
```

```
print(type(is_student))
```

```
<class 'bool'>
```

```
Q6.
What will be the output?

print(type("Python"))

A.

<class 'int'>

B.

<class 'float'>

C.

<class 'str'>

D.

<class 'bool'>

Ans: C
```

HW

```
What does the following code do?

name = input("Enter your name:")

A. Prints the name

B. Takes input from the user

C. Deletes the name

D. Creates an error
```

```
What data type is returned by input()?

A. int

B. float

C. bool

D. str
```

```
Which function converts a string to an integer?

A.

str()

B.

float()

C.

int()

D.
```

```
bool()
```